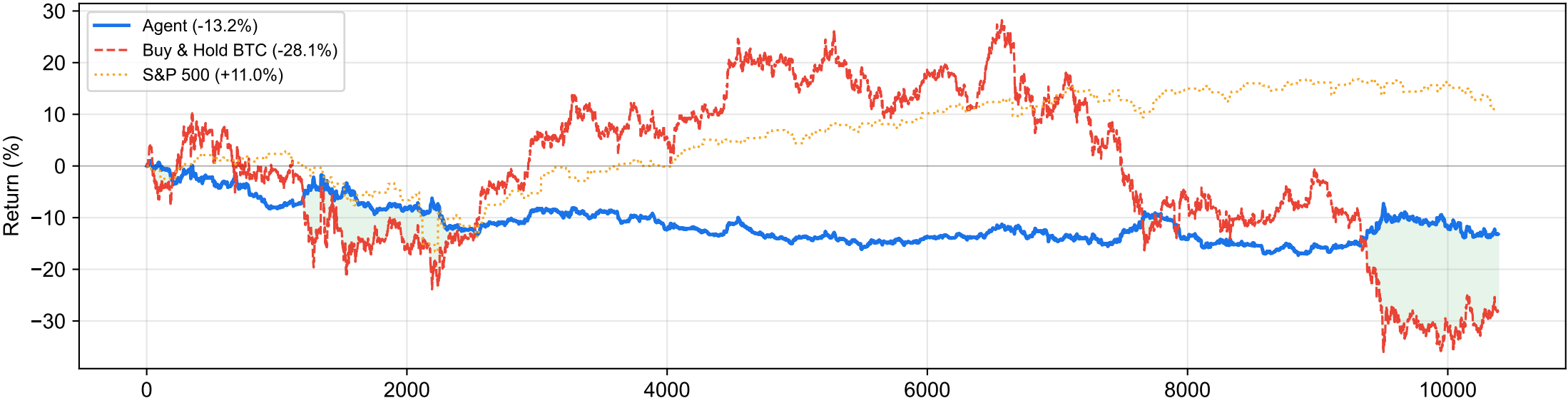
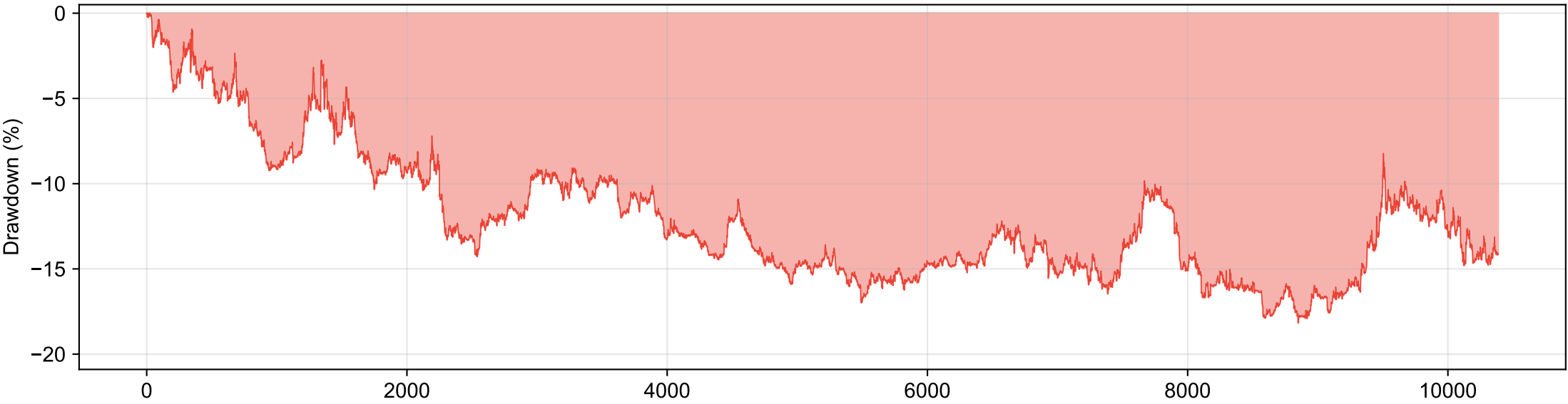


Evaluation Report: CNN+LSTM PPO × SPA (Pure OOS)
Senior AI Project — Out-of-Sample Results

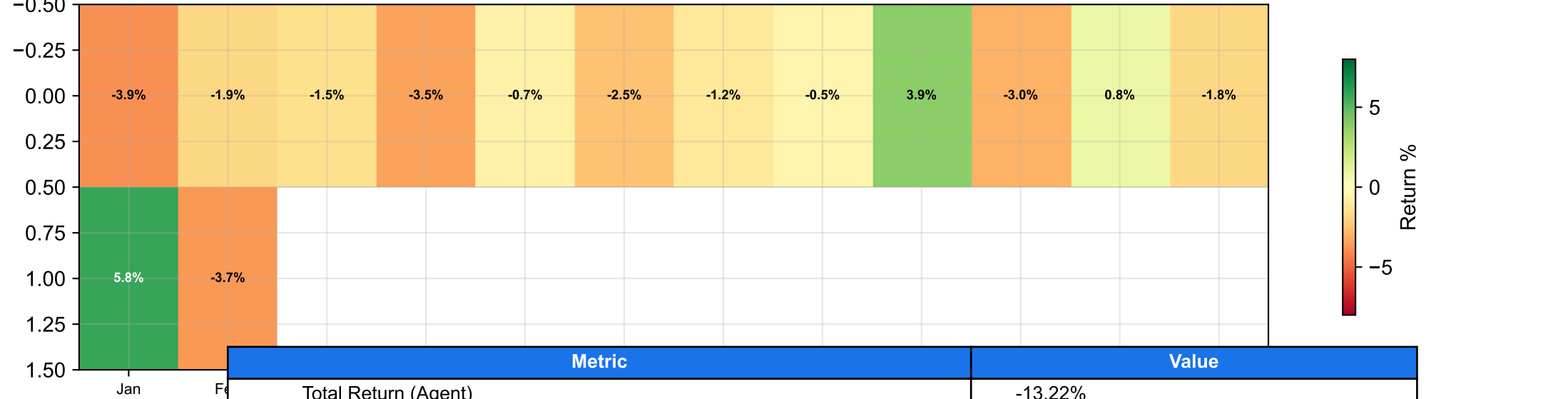
Cumulative Returns: Agent vs Buy & Hold & S&P 500



Drawdown (Underwater Plot)



Monthly Returns Heatmap



Metric		Value
Total Return (Agent)		-13.22%
Total Return (BTC Buy & Hold)		-28.10%
Alpha vs BTC B&H		+14.88%
Total Return (S&P 500)		+10.99%
Performance Summary		
Alpha vs S&P 500		-24.21%
Annualized Return		-11.27%
Annualized Volatility		13.71%
Sharpe Ratio (OOS)		-0.822
Sortino Ratio		-1.078
BENCHMARK SHARPE / SORTINO		
BTC Buy & Hold Sharpe		-0.527
BTC Buy & Hold Sortino		-0.671
S&P 500 Sharpe		0.517
S&P 500 Sortino		0.112
Max Drawdown		-18.17%
Win Rate		36.7%
Profit Factor		0.815
Total Trades		237
Avg Monthly Cashflow		\$348.69